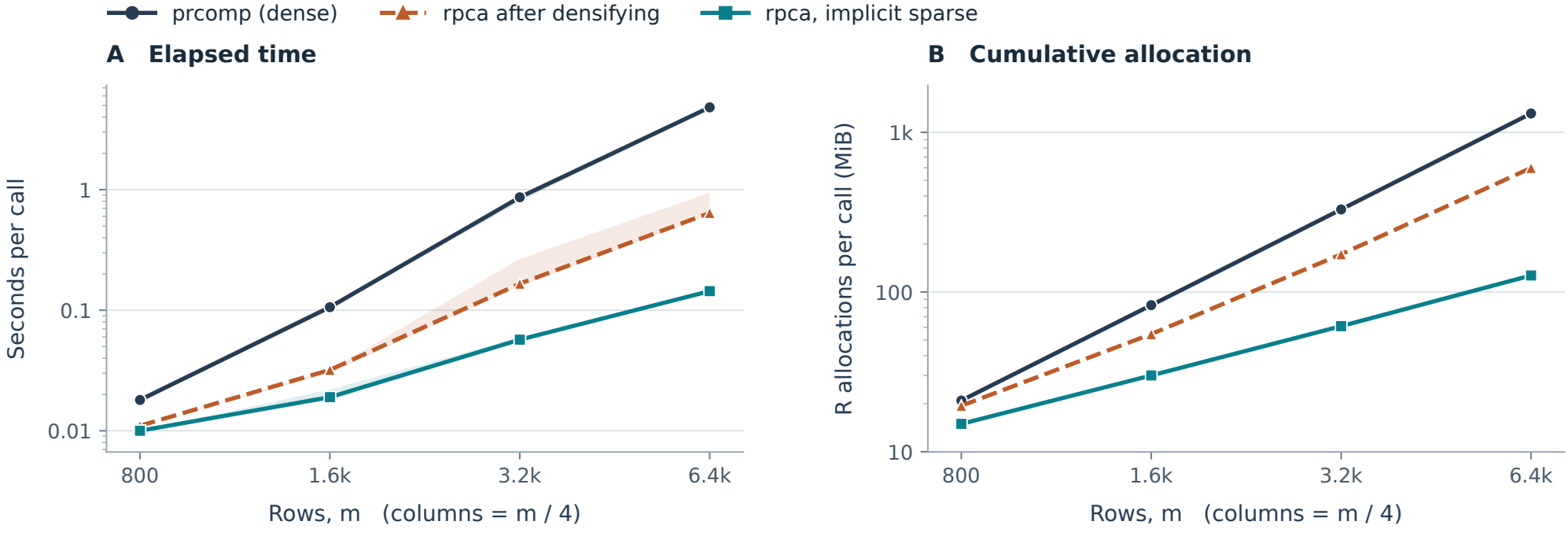
PCA on sparse input: preserving sparse storage

Same observations, centering, scaling, and 40 score columns; 5% density



k = 40, p = 10, q = 2 · timing: 5-run medians, IQR shaded · allocations: one profile · OpenBLAS, one thread

Largest observed error / optimal rank-40 error: 1.0001. R allocations exclude setup and do not measure peak RSS.